

CHAPTER 5

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5.1. Let $f : A \rightarrow B$ be an integral homomorphism of rings. Show that $f^* : \text{Spec } B \rightarrow \text{Spec } A$ is a closed mapping. (This is a geometric equivalent of (5.10).)

Proof. Take $\mathfrak{b} \leq B$, and recall that $f^*(V(\mathfrak{b})) \subset \overline{f^*(V(\mathfrak{b}))} = V(\mathfrak{b}^c)$ by (Ex. 1.21iii). Now, suppose $\mathfrak{p} \in V(\mathfrak{b}^c)$ is a prime ideal of A , i.e. $\mathfrak{p} \supset \mathfrak{b}^c = f^{-1}(\mathfrak{b})$. Then in particular $\mathfrak{p} \supset \ker f$, so $\mathfrak{p}$ descends to a prime ideal $f(\mathfrak{p})$ of $A/\ker f \cong \text{im } f \subset B$.

$\text{im } f \subset B$ is integral, so (5.10) gives us a prime $\mathfrak{q} \in \text{Spec } B$ with $\mathfrak{q} \cap \text{im } f = f(\mathfrak{p})$. We have that $\mathfrak{q} \supset f(\mathfrak{p}) \supset f(\mathfrak{b}^c)$ implies $\mathfrak{q} \supset \mathfrak{b}^{ce} = \mathfrak{b}$, i.e. $\mathfrak{q} \in V(\mathfrak{b})$. Further,

$$\begin{aligned} f^*(\mathfrak{q}) &= f^{-1}(\mathfrak{q}) \\ &= f^{-1}(\mathfrak{q} \cap \text{im } f) \\ &= f^{-1}(f(\mathfrak{p})) \\ &= \mathfrak{p}, \end{aligned}$$

since $\mathfrak{p} \supset \ker f$. Thus, $\mathfrak{p} \in f^*(V(\mathfrak{b}))$, so in fact $f^*(V(\mathfrak{b})) = \overline{f^*(V(\mathfrak{b}))}$ is closed. $\square$

5.2. Let B be integral over $A \subset B$, and let $f : A \rightarrow \Omega$ be a homomorphism of A into an algebraically closed field Ω . Show that f can be extended to a homomorphism of B into Ω .

Proof. $f : A \rightarrow \Omega$ corresponds to a maximal ideal $\mathfrak{p} = \ker f \leq A$ and an inclusion $\bar{f} : A/\mathfrak{p} \hookrightarrow \Omega$. (5.10) supplies us with a prime ideal $\mathfrak{q} \leq B$ such that $\mathfrak{q} \cap A = \mathfrak{p}$, which is in fact maximal by (5.8). Then $\pi : B \rightarrow B/\mathfrak{q}$ restricts to a map

$$\pi|_A : A \rightarrow \frac{A}{\mathfrak{q} \cap A} = \frac{A}{\mathfrak{p}} \subset \frac{B}{\mathfrak{q}}.$$

(5.6) gives us that $K := B/\mathfrak{q}$ is integral over $k := A/\mathfrak{p}$.

Let $(k, \bar{f}) \leq (C, \bar{g})$ be a maximal element of the poset Σ of maps from subrings of K to Σ , ordered by restriction. In particular, it is a valuation ring of K , so $K \supset C \supset \bar{k} = K$ by (5.22), whence $K = C$. Then $g := \bar{g} \circ \pi : B \rightarrow \Omega$ has

$$g|_A = \bar{g}|_k \circ \pi|_A = \bar{f} \circ \pi|_A = f.$$

$\square$

5.3. Let $f : B \rightarrow B'$ be a morphism of A -algebras, and let C be an A -algebra. If f is integral, prove that $f \otimes 1 : B \otimes_A C \rightarrow B' \otimes C$ is integral.

Proof. We have that B' is integral over $f(B) \leq B'$. That is, $f(B)[b'_1, \dots, b'_n]$ is finitely generated for all $b'_1, \dots, b'_n \in B'$. Note as well that $(f \otimes 1)(B \otimes C) \cong f(B) \otimes C$. Then, for arbitrary $\sum_{i=1}^n b'_i \otimes c_i \in B' \otimes_A C$,

$$(f \otimes 1)(B \otimes_A C) \left[\sum b'_i \otimes c_i \right] \leq f(B)[b'_1, \dots, b'_n] \otimes_A C[c_1, \dots, c_n].$$

The inclusions

$$f(B) \otimes_A C \hookrightarrow f(B)[b'_1, \dots, b'_n] \otimes_A C \hookrightarrow f(B)[b'_1, \dots, b'_n] \otimes_A C[c_1, \dots, c_n]$$

are finite by closure of finiteness under base extension (2.17), so the composition is finite by closure of finiteness under composition (2.16). Then our arbitrary element of $B' \otimes_A C$ is integral over $f(B) \otimes_A C$ by (5.1iii), so that $f \otimes 1$ is integral. $\square$

5.4 Let $A \subset B$ be a subring such that B integral over A . Let $\mathfrak{n}$ be a maximal ideal of B and let $\mathfrak{m} = \mathfrak{n} \cap A$ be the corresponding maximal ideal of A . Is $B_{\mathfrak{n}}$ necessarily integral over $A_{\mathfrak{m}}$?

Proof. We will show that this need not hold in general. Consider $A := k[x^2 - 1] \subset k[x] =: B$. Clearly $k[x]$ is integral over A , as it is a finitely generated A module. Consider the maximal ideal $\mathfrak{n} = (x - 1)$. We have that $\mathfrak{n} \supset (x^2 - 1)$ and $\mathfrak{m} := (x^2 - 1)^c$ is maximal in A , so $\mathfrak{n} \cap A \supset \mathfrak{m}$ with $\mathfrak{n}$ a proper ideal implies $\mathfrak{n} \cap A = \mathfrak{m}$.

Now, suppose that $1/(x + 1) \in k[x]_{(x-1)}$ is integral over $A_{\mathfrak{m}}$. Then we have $n \geq 0$ and $a_0, \dots, a_n \in A_{\mathfrak{m}}$ such that $(x + 1)^{-n-1} = a_n(x + 1)^{-n} + \dots + a_0$. Multiplying by $(x + 1)^{n+1}$, we get

$$1 = a_n(x + 1) + \dots + a_0(x + 1)^{n+1} \in (x + 1)k[x]_{(x+1)},$$

since $(x + 1) \cap k[x^2 - 1] = (x^2 - 1)$ by the same argument as above.

Note that $(x + 1)$ is exactly the kernel of the evaluation map $f : k[x] \rightarrow k : x \mapsto -1$. Then $k[x]_{(x+1)} = k[x]_S$ where $S = k[x] - \ker f$, so that $f(S) \subset k^\times$ gives us $k_{f(S)} \cong k$. Hence, we receive a map of localizations

$$k[x]_{(x+1)} \xrightarrow{f} k,$$

under which $1 \mapsto 1$ and $a_n(x + 1) + \dots + a_0(x + 1)^{n+1} \mapsto 0$. $1 \neq 0$ in k , so this contradicts our assumption. $\square$

5.7 Let A be a subring of B such that $B - A$ is closed under multiplication. Show that A is integrally closed in B .

Proof. Suppose that $b \in B - A$ is integral over A , so that $a_0 = a_1b + \dots + a_nb^n + b^{n+1}$ for $a_i \in A$ and minimal $n \geq 0$. In fact, we may assume $n > 0$, as otherwise $b \in A$. We have

$$a_0 = b(a_1 + \dots + a_nb^{n-1} + b^n) \in A$$

implies $a_1 + \dots + a_nb^{n-1} + b^n = a \in A$, as otherwise $a_0 \in B - A$ by closure. Then $a - a_1 = a_2b + \dots + a_nb^{n-1} + b^n$ is of strictly lower degree, contradicting minimality of n . $\square$

5.10. A ring homomorphism $f : A \rightarrow B$ is said to have the *going-up property* (resp. the *going-down property*) if the conclusion of the going-up theorem (resp. going-down theorem) holds for B and its subring $f(A)$. Let $f^* : \text{Spec } B \rightarrow \text{Spec } A$ be the mapping associated with f .

(a) Consider the following three statements:

- (i) f^* is a closed mapping.
- (ii) f has the going-up property.
- (iii) For any prime ideal $\mathfrak{q}$ of B , let $\mathfrak{p} = \mathfrak{q}^c$. Then $f^* : \text{Spec}(B/\mathfrak{q}) \rightarrow \text{Spec}(A/\mathfrak{p})$ is surjective.

Prove that (i) $\implies$ (ii) $\iff$ (iii).

(b) Consider the following three statements:

(i') f^* is an open mapping.

(ii') f has the going-down property.

(iii') For any prime ideal $\mathfrak{q}$ of B , let $\mathfrak{p} = \mathfrak{q}^c$. Then $f^* : \text{Spec}(B_{\mathfrak{q}}) \rightarrow \text{Spec}(A_{\mathfrak{p}})$ is surjective.

Prove that (i') $\implies$ (ii') $\iff$ (iii').

Proof. (a) We begin by showing that (ii) $\iff$ (iii). As in the proof of the going-up theorem, it suffices to consider the case of $\mathfrak{p} \subset \mathfrak{p}' \trianglelefteq A$ and $\mathfrak{q} \trianglelefteq B$ with $\mathfrak{q}^c = \mathfrak{p}$. Applying Spec to the following diagram,

$$\begin{array}{ccc} B & \xleftarrow{f} & A \\ \downarrow & & \downarrow \\ B/\mathfrak{q} & \xleftarrow{\bar{f}} & A/\mathfrak{p} \end{array}$$

we receive

$$\begin{array}{ccc} \text{Spec } B & \xrightarrow{f^*} & \text{Spec } A \\ \uparrow & & \uparrow \\ \text{Spec}(B/\mathfrak{q}) & \xrightarrow{f^*} & \text{Spec}(A/\mathfrak{p}). \end{array}$$

Using (1.21iii,iv) and $f^*(\mathfrak{q}) = \mathfrak{p}$, we can restrict to

$$\begin{array}{ccc} V(\mathfrak{q}) & \xrightarrow{f^*} & V(\mathfrak{p}) \\ \wr \uparrow & & \wr \uparrow \\ \text{Spec}(B/\mathfrak{q}) & \xrightarrow{f^*} & \text{Spec}(A/\mathfrak{p}). \end{array}$$

Then there exists $\mathfrak{q}' \supset \mathfrak{q}$ with $f^{-1}(\mathfrak{q}') = \mathfrak{p}'$ if and only if $\bar{\mathfrak{p}}' \in \text{im } f^*$. Hence, surjectivity of $f^* : \text{Spec}(B/\mathfrak{q}) \rightarrow \text{Spec}(A/\mathfrak{p})$ holds for all prime ideals $\mathfrak{q} \trianglelefteq B$ if and only if $f : A \rightarrow B$ possesses the going-up property.

Now, we will show that (i) $\implies$ (iii). Suppose f^* is a closed mapping, and recall the last diagram from above. By (1.21iii) once more, we have that

$$\begin{aligned} f^*(V(\mathfrak{q})) &= \overline{f^*(V(\mathfrak{q}))} \\ &= V(\mathfrak{q}^c) \\ &= V(\mathfrak{p}), \end{aligned}$$

giving surjectivity of $f^* : \text{Spec}(B/\mathfrak{q}) \rightarrow \text{Spec}(A/\mathfrak{p})$.

(b) We begin by showing that (ii') $\iff$ (iii'). Once again, we need only consider the case of prime ideals $\mathfrak{p}' \subset \mathfrak{p} \trianglelefteq A$ with $\mathfrak{q} \trianglelefteq B$ such that $\mathfrak{q}^c = \mathfrak{p}$. Fixing $\mathfrak{q}$ and $\mathfrak{p} = \mathfrak{q}^c$, (3.11iv) gives us the diagram

$$\begin{array}{ccc} \{\mathfrak{q}' \subset \mathfrak{q}\} & \xrightarrow{f^*} & \{\mathfrak{p}' \subset \mathfrak{p}\} \\ \wr \uparrow & & \wr \uparrow \\ \text{Spec } B_{\mathfrak{q}} & \xrightarrow{f^*} & \text{Spec } A_{\mathfrak{p}}, \end{array}$$

whence we conclude our result similarly.

Finally, we show that (i') $\implies$ (ii'). Suppose $f^* : \text{Spec } B \rightarrow \text{Spec } A$ is open. Fix $\mathfrak{p}' \subset \mathfrak{p} \trianglelefteq A$, and $\mathfrak{q} \trianglelefteq B$ lying over $\mathfrak{p}$, i.e. with $\mathfrak{q}^c = \mathfrak{p}$. For any $b \in B \setminus \mathfrak{q}$, $f^*(X_b) \ni \mathfrak{p}$ is open, so $\mathfrak{p}' \rightsquigarrow \mathfrak{p}$ implies $\mathfrak{p}' \in f^*(X_b)$. This supplies us with $\mathfrak{q}'_b \in X_b$ such that $f^*(\mathfrak{q}'_b) = \mathfrak{p}'$, whence we receive

$$\mathfrak{q}' := \bigcap_{b \in B \setminus \mathfrak{q}} \mathfrak{q}'_b \subset \mathfrak{q}$$

such that

$$f^*(\mathfrak{q}') = \bigcap_{b \in B \setminus \mathfrak{q}} f^*(\mathfrak{q}'_b) = \mathfrak{p}',$$

as desired. $\square$

5.11 Let $f : A \rightarrow B$ be a flat homomorphism of rings. Then f has the going down property.

Proof. This is a corollary of (Ex. 3.18) and (Ex. 5.10b). $\square$

5.12 Let G be a finite group of automorphisms of a ring A , and let A^G denote the subring of G -invariants. Prove that A is integral over A^G .

Let S be a multiplicatively closed subset of A such that $\sigma(S) \subset S$ for all $\sigma \in G$, and let $S^G = S \cap A^G$. Show that the action of G on A extends to an action on $S^{-1}A$, and that $(S^G)^{-1}A^G \cong (S^{-1}A)^G$.

Proof. Fix $x \in A$, and consider the monic polynomial $f(t) = \prod_{\sigma \in G} (t - \sigma(x)) \in A[t]$. We have

$$\begin{aligned} \tau(f(t)) &= \prod_{\sigma \in G} \tau(t - \sigma(x)) \\ &= \prod_{\sigma \in G} (t - \tau\sigma(x)) \\ &= f(t), \end{aligned}$$

so that $f(t) \in A[t]^G = A^G[t]$. Since $f(x) = 0$, we're done.

We define the G action on $S^{-1}A$ by $\sigma(a/s) = \sigma(a)/\sigma(s)$. $\sigma(s) \in \sigma(S) \subset S$ by assumption. Now, fix $\sigma \in G$ and let $a/s = a'/s'$. We have $s'' \in S$ with $s''(as' - a's) = 0$, so that

$$0 = \sigma(s'')(\sigma(a)\sigma(s') - \sigma(a')\sigma(s))$$

for $\sigma(s'') \in S$ implies $\sigma(a/s) = \sigma(a'/s')$. We conclude that the action is well-defined, and clearly coincides on $\text{im}(A \hookrightarrow S^{-1}A)$.

If $a \in A^G$, then $\sigma(a/1) = a/1$. Thus, the natural map $\phi(a) = a/1$ restricts to an injection $\phi : A^G \hookrightarrow (S^{-1}A)^G$. We wish to show that ϕ satisfies the conditions of Corollary (3.2).

(1) If $s \in S^G$, then

$$\sigma(1/s) = 1/\sigma(s) = 1/s \in (S^{-1}A)^G.$$

(2) Suppose now that $\phi(a) = 0$ for $a \in A^G$. We have some $s \in S$ such that $sa = 0$. Then $s' = \prod_{\sigma \in G} \sigma(s) \in S^G$ has $s'a = 0$, as desired.

(3) Let $a/s \in (S^{-1}A)^G$. That is, we can find $s_\sigma \in S$ for each $\sigma \in G$ such that

$$s_\sigma(\sigma(a)s - a\sigma(s)) = 0.$$

Taking

$$s' := \prod_{\sigma, \tau \in G} \sigma(s_\tau)$$

gives us $s' \in S^G$ such that

$$s'(\sigma(a)s - a\sigma(s)) = 0$$

for all $\sigma \in G$.

Suppose now that $ss' \neq \sigma(ss') = \sigma(s)s'$ □

5.13 In the situation of (Ex. 5.12), let $\mathfrak{p}$ be a prime ideal of A^G , and let P be the set of prime ideals of A whose contraction is $\mathfrak{p}$. Show that G acts transitively on P . In particular, P is *finite*.

Proof. Let $\mathfrak{q}, \mathfrak{q}' \in P$ be primes lying over $\mathfrak{p}$, and let $\sigma \in G$. $\mathfrak{q}$ is prime and σ is an isomorphism, so $\sigma(\mathfrak{q}) = (\sigma^{-1})^{-1}(\mathfrak{q})$ is prime. Let $x' \in \mathfrak{q}'$. Then

$$x := \prod_{\sigma \in G} \sigma^{-1}(x') \in \mathfrak{p} \subset \mathfrak{q}',$$

so there is some $\sigma \in G$ such that $x' = \sigma(x) \in \sigma(\mathfrak{q})$. This gives us

$$\mathfrak{q}' \subset \bigcup_{\sigma \in G} \sigma(\mathfrak{q}),$$

where each $\sigma(\mathfrak{q})$ is prime. (1.11) gives us $\sigma \in G$ such that $\mathfrak{q}' \subset \sigma(\mathfrak{q})$, whence (5.9) and (Ex. 5.12) give us $\mathfrak{q}' = \sigma(\mathfrak{q})$. □

5.14 Let A be an integrally closed domain, K its field of fractions and L a finite normal separable extension of K . Let G be the Galois group of L over K and let B be the integral closure of A in L . Show that $\sigma(B) = B$ for all $\sigma \in G$, and that $A = B^G$.

Proof. Let $x \in B$ have minimal polynomial $f(t) \in A[t]$, let $N := \text{Stab}_G(x)$, and consider the monic polynomial

$$g(t) = \prod_{\sigma N \in G/N} (t - \sigma(x)).$$

Clearly $g(x) = 0$. We have $f(\sigma(x)) = \sigma(f(x)) = 0$, since G fixes $A \subset K$. Thus,

$$\deg f \geq |G|/|\text{Stab}_G(x)|$$

□

5.15 Let A, K as in (Ex. 5.14), let L be any finite extension field of K , and let B be the integral closure of A in L . Show that, if $\mathfrak{p}$ is any prime ideal of A , then the set of prime ideals $\mathfrak{q}$ of B which contract to $\mathfrak{p}$ is finite (in other words, that $\text{Spec } B \rightarrow \text{Spec } A$ has *finite fibers*).

Proof. Let $G = \text{Gal}(L/K)$. (5.14) gives us that $A = B^G$, whence the result follows directly from (5.13). □

5.16. Let k be a field and let $A \neq 0$ be a finitely generated k -algebra. Then there exist elements $y_1, \dots, y_r \in A$ which are algebraically independent over k and such that A is integral over $k[y_1, \dots, y_r]$.

We shall assume that k is *infinite*. (A different proof is required for the finite case.) From the proof, it follows that $y_1, \dots, y_r$ may be chosen to be linear combinations

of $x_1, \dots, x_n$. This has the following geometrical interpretation: if k is algebraically closed and X is an affine algebraic variety in k^n with coordinate ring $A \neq 0$, then there exists a linear subspace of dimension r in k^n and a linear mapping of k^n onto L which maps X onto L . [Use Exercise (2).]

5.27 Let A, B be two local rings. B is said to *dominate* A if A is a subring of B and the maximal ideal $\mathfrak{m}$ of A is contained in the maximal ideal $\mathfrak{n}$ of B (or, equivalently, if $\mathfrak{m} = \mathfrak{n} \cap A$).

Let K be a field and let Σ be the set of all local subrings of K . If Σ is ordered by the relation of domination, show that Σ has maximal elements, and that $A \in \Sigma$ is maximal if and only if A is a valuation ring of K .

Proof. Fix a field K . We will begin by showing that Σ has maximal elements. By Zorn's lemma, it suffices to show that a chain

$$\begin{array}{ccccccc} A_1 & \leq & A_2 & \leq & \cdots \\ \cup & & \cup & & \\ \mathfrak{m}_1 & \subset & \mathfrak{m}_2 & \subset & \cdots \end{array}$$

must have an upper bound in Σ . We consider the ring $A := \bigcup A_i$. Because each A_{i+1} dominates A_i , the $\mathfrak{m}_i$ form an increasing chain of subrings of A . Their union $\mathfrak{m} := \bigcup \mathfrak{m}_i$ is an ideal, as $f \in A_i$, $x \in \mathfrak{m}_j$ implies $fx \in \mathfrak{m}_{i+j}$. If $f \in A \setminus \mathfrak{m}$, then $f \in A_i - \mathfrak{m}_i = A_i^\times$ is a unit, whence A is a local ring with unique maximal ideal $\mathfrak{m}$ by. Then $\mathfrak{m} \cap A_i = \mathfrak{m}_i$ gives us $A_i \leq A$ for all i , so that $A \in \Sigma$ is in fact an upper bound to our chain.

Suppose that $A < B \in \Sigma$ have ideals $\mathfrak{m} \leq A$ and $\mathfrak{n} \leq B$. It suffices to show that A is not a valuation ring. Take $f \in B \setminus A$ supplied by $B \neq A$. We have $f^{-1} \notin A$, as otherwise $f^{-1} \in \mathfrak{m} \setminus \mathfrak{n}$ contradicts $A \leq B$. Thus, A is not a valuation ring.

Let $A \in \Sigma$ be maximal, and suppose it is not a valuation ring. We can take $x \in K$ such that $x, x^{-1} \notin A$. Then $(\mathfrak{m}, x) \leq A[x]$ is a proper ideal of $A \subsetneq A[x]$, and is thus contained in some maximal ideal $\mathfrak{n} \leq A[x]$. Localizing at $\mathfrak{n}$, we receive a local ring $A' := A[x]_{\mathfrak{n}} \supset A$ with unique maximal ideal $\mathfrak{n}' := \mathfrak{n}A' \supset \mathfrak{m}$. Hence, A' dominates A , contradicting maximality of A . We conclude that A is indeed a valuation ring. $\square$

5.28 Let A be an integral domain, K its field of fractions. Show that the following are equivalent:

- (a) A is a valuation ring of K ;
- (b) If $\mathfrak{a}, \mathfrak{b}$ are any two ideals of A , then either $\mathfrak{a} \subset \mathfrak{b}$ or $\mathfrak{b} \subset \mathfrak{a}$.

Deduce that if A is a valuation ring and $\mathfrak{p}$ is a prime ideal of A , then $A_{\mathfrak{p}}$ and $A/\mathfrak{p}$ are valuation rings of their fields of fractions.

Proof. ((a) $\implies$ (b)) Let $\mathfrak{a}, \mathfrak{b} \leq A$, and suppose without loss of generality that $\mathfrak{a} \not\subset \mathfrak{b}$. Then fix $f \in \mathfrak{a} \setminus \mathfrak{b}$ and let $0 \neq g \in \mathfrak{b}$. $f/g \notin A$, so A a valuation ring gives $g/f \in A$, whence $g \in \mathfrak{a}$, so that $\mathfrak{b} \subset \mathfrak{a}$.

((b) $\implies$ (a)) Let $x/y \in K$, and consider $(x), (y) \leq A$. If $(x) \subset (y)$, then $x \in (y)$ implies $x/y \in A$. If $(y) \subset (x)$, then $y \in (x)$ implies $y/x \in A$.

Now, suppose A is a valuation ring and $\mathfrak{p} \leq A$ is a prime ideal. Any $\mathfrak{a}, \mathfrak{b} \leq A_{\mathfrak{p}}$ are extended, and $\mathfrak{a}^c \subset \mathfrak{b}^c$ or $\mathfrak{a}^c \supset \mathfrak{b}^c$ by the above, whence $\mathfrak{a} \subset \mathfrak{b}$ or $\mathfrak{a} \supset \mathfrak{b}$. The ideal lattice of $A/\mathfrak{p}$ is isomorphic to the poset $\{\mathfrak{a} \supset \mathfrak{p}\}$, so it is clear that the second criterion holds. $\square$

5.29 Let A be a valuation ring of a field K . Show that every subring of K which contains A is a localization of A .

Proof. Let B be a subring of K containing A . By (5.18i,ii), B is a valuation ring, and hence local with maximal ideal $\mathfrak{m} \triangleleft B$. Let $\mathfrak{n} := A \cap \mathfrak{m}$ be the contraction of $\mathfrak{m}$ by $A \hookrightarrow B$, and consider the localization $A_{\mathfrak{n}} \subset B$. If $f \in B \setminus A$, then $f^{-1} \in A$, so that $f \in B^{\times} = B \setminus \mathfrak{m}$ implies $f = (f^{-1})^{-1} \in A_{\mathfrak{n}}$, whence $B \subset A_{\mathfrak{n}}$. $\square$

5.30 Let A be a valuation ring of a field K . The group $A^{\times}$ of units of A is a subgroup of the multiplicative group $K^{\times}$ of K .

Let $\Gamma = K^{\times}/A^{\times}$. If $\xi, \eta \in \Gamma$ are represented by $x, y \in K$, define $\xi \geq \eta$ to mean $xy^{-1} \in A$. Show that this defines a total ordering on Γ which is compatible with the group structure (i.e., $\xi \geq \eta \implies \xi\omega \geq \eta\omega$ for all $\omega \in \Gamma$). In other words, Γ is a totally ordered abelian group. It is called the *value group* of A .

Let $v : K^{\times} \rightarrow \Gamma$ be the canonical homomorphism. Show that $v(x + y) \geq \min(v(x), v(y))$ for all $x, y \in K^{\times}$.

Proof. Clearly $\geq$ is reflexive and transitive. If $\xi \geq \eta$ and $\eta \geq \xi$, then $xy^{-1} \in A$ and $yx^{-1} \in A$ implies $xy^{-1} \in A^{\times}$, so that $\xi = \eta$ gives us anti-symmetry. It is total by definition of a valuation ring. Now, we show it is compatible with the group structure. Let $\xi \geq \eta$, $\omega \in \Gamma$, and take representatives $x, y, z \in K$. Then $(xz)(yz)^{-1} = xy^{-1} \in A$ implies $\xi\omega \geq \eta\omega$.

Suppose, without loss of generality, that $v(x) \geq v(y)$, i.e. $xy^{-1} \in A$. Then

$$(x + y)y^{-1} = xy^{-1} + 1 \in A$$

implies $v(x + y) \geq v(y)$, as desired. Then $d(x, y) := -v(x - y)$ defines an ultrametric on K , thus giving K the structure of a non-archimedean field. $\square$

5.31 Conversely, let Γ be a totally ordered abelian group (written *additively*), and let K be a field. A *valuation of K with values in Γ* is a mapping $v : K^{\times} \rightarrow \Gamma$ such that

- (a) $v(xy) = v(x) + v(y)$,
- (b) $v(x + y) \geq \min(v(x), v(y))$,

for all $x, y \in K^{\times}$. Show that the set of elements $x \in K$ such that $x = 0$ or $v(x) \geq 0$ is a valuation ring of K . This ring is called the valuation ring of v , and the subgroup $v(K^{\times})$ is the *value group* of v .

Thus, the concepts of valuation ring and valuation are essentially equivalent.

Proof. Define $A := \{x \in K : x = 0 \text{ or } v(x) \geq 0\}$. Then $0 \in A$, A is additive by (b), and multiplicative by (a). If $x \in K^{\times}$, then

$$v(x) + v(x^{-1}) = v(1) = 0$$

implies either $v(x) \geq 0$ or $v(x^{-1}) \geq 0$. That is, $x \in A$ or $x^{-1} \in A$. We conclude that A is indeed a valuation ring of K . $\square$

5.32 Let Γ be a totally ordered abelian group. A subgroup $\Delta \leq \Gamma$ is *isolated* in Γ if, whenever $0 \leq \beta \leq \alpha$ and $\alpha \in \Delta$, we have $\beta \in \Delta$. Let A be a valuation ring of a field K , with value group Γ . If $\mathfrak{p}$ is a prime ideal of A , show that $v(A \setminus \mathfrak{p})$ is the set of elements ≥ 0 in an isolated subgroup $\Delta \leq \Gamma$, and that the mapping so defined of $\text{Spec } A$ into isolated subgroups of Γ is surjective.

If $\mathfrak{p} \triangleleft A$ is prime, what are the value groups of the valuation rings $A/\mathfrak{p}$, $A_{\mathfrak{p}}$?

Proof. Suppose $\alpha = v(a) \in v(A \setminus \mathfrak{p})$, and let $0 \leq \beta \leq \alpha$ for $\beta = v(b) \in v(A)$. Then $ab^{-1} \in A$, so $b \in A \setminus \mathfrak{p}$, as otherwise $a = (ab^{-1})b \in \mathfrak{p}$. Hence, $\beta \in v(A \setminus \mathfrak{p})$.

Now, let $\Delta \leq \Gamma$ be an isolated subgroup, and consider $\mathfrak{p} := v^{-1}(\Gamma \setminus \Delta) \cup \{0\}$. Additivity is clear. If $b \in v^{-1}(\Gamma \setminus \Delta)$ and $a \in A \setminus \{0\}$, then $v(ab) = v(a) + v(b) \geq v(b) \geq 0$, so $v(ab) \notin \Delta$ gives us that $\mathfrak{p}$ is an ideal. Finally, if $a, b \in A \setminus \mathfrak{p}$, then $v(ab) = v(a) + v(b) \in \Delta$ implies $ab \in A \setminus \mathfrak{p}$, whence $\mathfrak{p}$ is prime. It is clear that $\mathfrak{p} \mapsto \Gamma$ under the mapping of $\text{Spec } A$.

The value group of $A_{\mathfrak{p}}$ is

$$\begin{aligned} \frac{K^\times}{A_{\mathfrak{p}}^\times} &= \frac{K^\times}{A_{\mathfrak{p}} \setminus \mathfrak{p}A_{\mathfrak{p}}} \\ &= \frac{K^\times / A^\times}{(A_{\mathfrak{p}} \setminus \mathfrak{p}A_{\mathfrak{p}}) / A^\times} \\ &= \frac{\Gamma}{\pm v(A \setminus \mathfrak{p})} \\ &= \Gamma / \Delta. \end{aligned}$$

Dually, the value group of $A/\mathfrak{p}$ is

$$\frac{\text{Frac}(A/\mathfrak{p})}{(A/\mathfrak{p})^\times} =$$

□

5.33 Let Γ be a totally ordered abelian group, k any field, and $A := k[\Gamma]$ the group algebra of Γ over k . Show that A is an integral domain.

If $u = \lambda_1 \alpha_1 + \cdots + \lambda_n \alpha_n$ is any nonzero element of A , where each $\lambda_i \neq 0$ and $\alpha_1 < \cdots < \alpha_n$, define $v_0(u) := \alpha_1$.¹ Show that $v_0 : A \setminus \{0\} \rightarrow \Gamma$ satisfies conditions (a) and (b) of (Ex. 5.31).

Let K be the fraction field of A . Show that v_0 can be uniquely extended to a valuation v of K , and that the value group of v is precisely Γ .

Proof. Take $u, u' \in A$ nonzero and write

$$\begin{aligned} u &= \lambda_1 x_{\alpha_1} + \cdots + \lambda_m x_{\alpha_m}, \\ u' &= \lambda'_1 x_{\alpha'_1} + \cdots + \lambda'_n x_{\alpha'_n} \end{aligned}$$

for $\lambda_i, \lambda'_i \neq 0$ and $\alpha_1 < \cdots < \alpha_m, \alpha'_1 < \cdots < \alpha'_n$. Suppose by way of contradiction that

$$\begin{aligned} 0 &= uu' \\ &= \sum_{i=1}^m \sum_{j=1}^n \lambda_i \lambda'_j x_{\alpha_i + \alpha'_j}. \end{aligned}$$

Then $\alpha_1 + \alpha'_1 < \alpha_i + \alpha'_j$ for all $i, j \neq 1$, so $\lambda_1 \lambda'_1 x_{\alpha_1 + \alpha'_1} = 0$ implies $\lambda_1 = 0$ or $\lambda'_1 = 0$, a contradiction.

Let $u, u' \in A$ nonzero. By the above, $uu' = \lambda_1 \lambda'_1 x_{\alpha_1 + \alpha'_1} + \cdots$, so $v_0(uu') = \alpha_1 + \alpha'_1$ gives us (a). For (b), we allow $m, n = 0$ and induct on $k = \#\{i \mid \lambda_i + \lambda'_i = 0, \alpha_i = \alpha'_i\}$. If $k = 0$, then clearly $v_0(u + u') = \min\{v_0(u), v_0(u')\}$. If $k = m = n$, then $u + u' = 0$, so the result holds trivially. If only one of $m, n = 0$, then without loss of generality $n = 0$, so

¹This arises from the natural Γ -graded structure of $k[\Gamma]$.

$v_0(u + u') = v_0(u) = \min(v_0(u), \infty)$. Otherwise, $v_0(u + u') = v_0(u'' + u''')$ for $u'' = u - \lambda_1 x_{\alpha_1}$, $u''' = u' - \lambda_1 x_{\alpha_1}$, and $k' = k - 1$, so

$$\begin{aligned} v_0(u + u') &= v_0(u'' + u''') \\ &\geq \min(v_0(u''), v_0(u''')) \\ &\geq \min(v_0(u), v_0(u')), \end{aligned}$$

whence the result follows by induction.

Take $K = \text{Frac } A$. Then for any $x, y \in A$ and $y \neq 0$,

$$v(xy^{-1}) = v((xy^{-1})y) = v(xy^{-1}) + v(y)$$

forces $v(xy^{-1}) = v_0(x) - v_0(y)$. Clearly this is a valuation on K , and $v : K^\times \rightarrow \Gamma$ is surjective by construction, so Γ is the valuation group of v . $\square$

5.34 Let A be a valuation ring and K its field of fractions. Let $f : A \rightarrow B$ be a ring homomorphism such that $f^* : \text{Spec } B \rightarrow \text{Spec } A$ is a closed mapping. Then if $g : B \rightarrow K$ is any A -algebra homomorphism (i.e., $g \circ f$ is the embedding $A \hookrightarrow K$), we have $g(B) = A$.

Proof. Set $C := g(B)$. g an A -algebra homomorphism gives us $A \subset C \subset K$, whence C is a valuation ring by (5.18ii), and both A, C are local by (5.18i). Let $\mathfrak{m} \trianglelefteq A$ and $\mathfrak{n} \trianglelefteq C$ be their respective maximal ideals. f is closed by hypothesis, while $g : B \rightarrow C$ is surjective, and thus a closed embedding. Then the embedding $A \hookrightarrow C$ is closed, so that $\mathfrak{n}^c$ is maximal. We conclude that $\mathfrak{n}^c = \mathfrak{m}$, so that C dominates A , meaning $C = A$ by maximality of valuation rings under domination (Ex. 5.27). Alternatively, (Ex. 5.29) gives us $C = A_{\mathfrak{p}}$, but $\mathfrak{p} = \mathfrak{p}A_{\mathfrak{p}}^c = \mathfrak{n}^c = \mathfrak{m}$ implies $C = A_{\mathfrak{m}} = A$. $\square$

5.35 From (Ex. 5.1) and (Ex. 5.3), it follows that, if $f : A \rightarrow B$ is integral and C is any A -algebra, then the mapping $(f \otimes 1)^* : \text{Spec}(B \otimes_A C) \rightarrow \text{Spec } C$ is a closed map.

Conversely, suppose that $f : A \rightarrow B$ has this property and that B is an integral domain. Then f is integral.

Show that the result just proved remains valid if B is a ring with only finitely many minimal prime ideals (e.g., if B is Noetherian).

Proof. Recall that $f : A \rightarrow B$ is said to be integral if $f(A) \subset B$ is integral. Let $\pi : A \rightarrow A/\ker f$ be the projection and $\bar{f} : A/\ker f \hookrightarrow B$ the map with $f = \bar{f} \circ \pi$. Right exactness of tensor products gives us that $\pi \otimes 1 : C \rightarrow (A/\ker f) \otimes_A C$ is surjective. Then $(\pi \otimes 1)^*$ is a closed embedding, so

$$(f \otimes 1)^* = (\pi \otimes 1)^* \circ (\bar{f} \otimes 1)$$

implies

$$\begin{aligned} (\bar{f} \otimes 1)^*(U) &= (\pi \otimes 1)^{*^{-1}}((\pi \otimes 1)^* \circ (\bar{f} \otimes 1)) \\ &= (\pi \otimes 1)^{*^{-1}}((f \otimes 1)^*(U)), \end{aligned}$$

which is closed whenever U is closed and f satisfies our property. Thus, $\bar{f} : A/\ker f \hookrightarrow B$ satisfies our property, so we may reduce to the case of $A \cong f(A) \subset B$.

Suppose first that $A \subset B$ are integral domains, B has fraction field K , and that the inclusion $f : A \hookrightarrow B$ has the above property. We wish to show that the integral closure of

A in B is precisely B . It suffices to show that the integral closure $\overline{A}$ contains B . By (5.22), it suffices to show that $B \subset C$ for all valuation rings $A \subset C \subset K$.

Take $A \subset C \subset K$ to be an arbitrary valuation ring of K containing A . Consider the maps $f \otimes 1 : C \rightarrow B \otimes_A C$ and $g : B \otimes_A C \rightarrow K : b \otimes c \mapsto bc$. We have that $g \circ (f \otimes 1) : C \rightarrow K$ is the embedding, and $(f \otimes 1)^* : \text{Spec } B \otimes_A C \rightarrow \text{Spec } C$ is closed. Thus, (Ex. 5.34) gives us $g(B \otimes_A C) = C$, but $B \subset g(B \otimes_A C)$, so $B \subset C$ as desired.

Now, take B with finitely many minimal prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n \trianglelefteq B$ and subring $A \subset B$ such that $f : A \rightarrow B$ has the above property. Then each $A \rightarrow B \rightarrow B/\mathfrak{p}_i$ has the above property since each $\pi_i \otimes 1 : B \otimes_A C \rightarrow (B/\mathfrak{p}_i) \otimes_A C$ is surjective and tensor products are functorial. Thus, the $A \rightarrow B/\mathfrak{p}_i$ are integral, whence $A \rightarrow \prod B/\mathfrak{p}_i$ is integral by (Ex. 5.6).

The map $\prod B/\mathfrak{p}_i \rightarrow B/\sqrt{0}$ given by $(x_i + \mathfrak{p}_i)_{i=1}^n \mapsto \sum_i x_i + \sqrt{0}$ is surjective, so $A \rightarrow B/\sqrt{0} := \overline{B}$ is integral. It suffices, then, to prove the following lemma.

Lemma 1. *If $\mathfrak{b} \trianglelefteq B$ is an ideal such that $b \in \mathfrak{b}$ implies $b^n \in A$ for some $n > 0$, and $A/\mathfrak{b}^c \rightarrow B/\mathfrak{b}$ is integral, then $A \rightarrow B$ is integral. In particular, if $A \rightarrow B/\sqrt{0}$ is integral, then $A \rightarrow B$ is integral.*

Proof. Define $\overline{A} := A/\mathfrak{b}^c$ and $\overline{B} := B/\mathfrak{b}$. Then, for each $x \in B$, we have a monic polynomial $\overline{f}(t) \in \overline{A}[t]$ such that $\overline{f}(\overline{x}) = \overline{0}$. Take $f(t) \in A[t]$ which projects to $\overline{f}$. Then $f(x) \in \mathfrak{b}$, so there exists $n > 0$ such that $f(x)^n \in A$. $f(t)$ is monic of positive degree, so $f(t)^n \in A[t]$ is monic of positive degree, whence $f(t)^n - f(x)^n \in A[t]$ is as well. Clearly x is a root of $f(t)^n - f(x)^n$, so that x is integral over A , as desired. $\square$

We conclude that $A \rightarrow B$ is integral, completing our proof. $\square$